Ethics, Privacy, and Security in Data Science and AI (CST4235)

Week 3

(15 Credits)

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Privacy-Preserving AI

Today's Agenda

- What is privacy why do we need private ML algorithms
- Four pillars of Perfectly-Privacy Preserving AI
- How we can satisfy all four pillars
- Differential privacy in ML
- Differential privacy in practice



Privacy ≠ Anonymisation

Anonymisation alone **does not guarantee privacy**.
Even when personal identifiers (like names or IDs) are removed, data can often be re-identified by linking it with other datasets.

Privacy means protecting individuals' information even when data is analysed, shared, or combined, not just removing names.



Examples:

- The **Netflix Prize dataset (2006)** was anonymised, yet researchers re-identified users by comparing movie ratings with public IMDb profiles.
- **Health data** stripped of identifiers can still reveal individuals when combined with demographic or location information.
- **Facial images** blurred for anonymity can sometimes be reconstructed using AI models trained on similar datasets.

Important Points:

- Anonymisation is one *technique* for privacy, but not a *complete solution*.
- True privacy-preserving AI requires methods such as **differential privacy**, **federated learning**, and **secure computation** to protect individuals even in complex data

Machine Learning Models Are Vulnerable

AI models are like that one friend who promises to keep a secret...
but then accidentally blurts it out in a group chat.

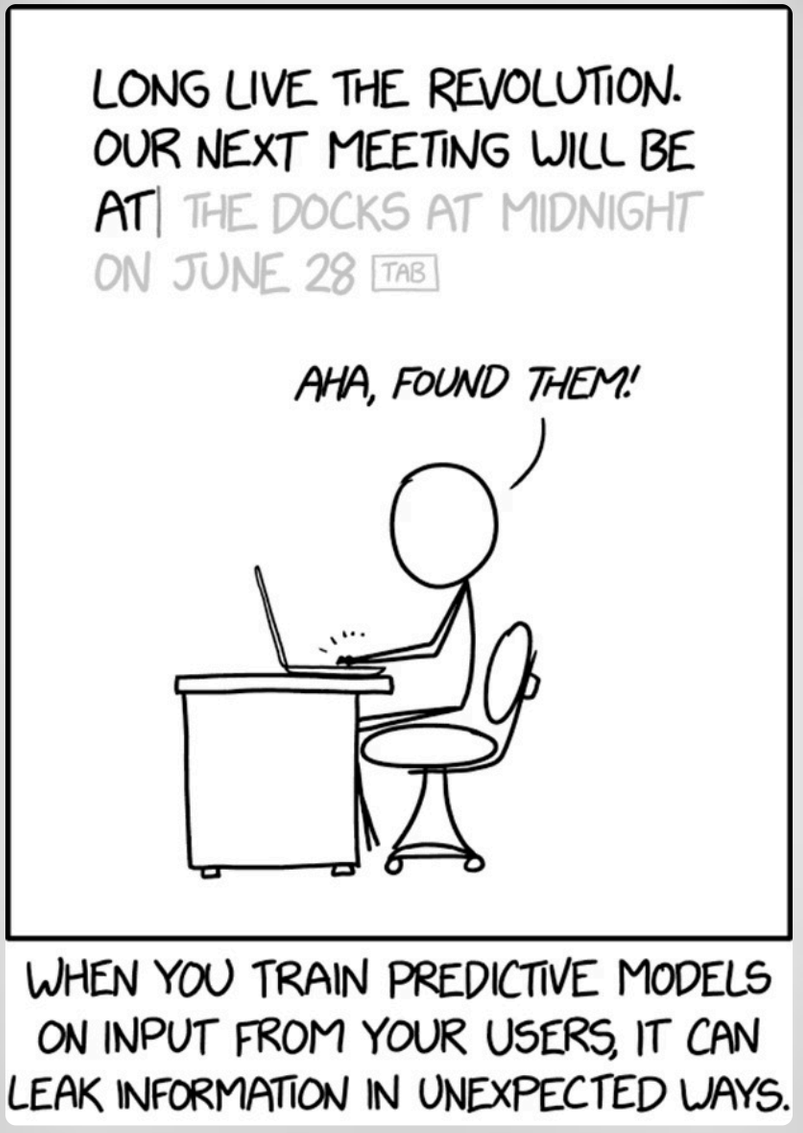
When we train models on sensitive data, they sometimes "remember" things they should not –
from phone numbers and passwords to secret revolution plans.

Examples:

- A language model trained on user chat data accidentally starts quoting real conversations.
- A facial recognition model "recalls" someone who was supposedly anonymised.
- A health model leaks patient information by overfitting to unique medical patterns.

The moral of the Story

What happens in training should stay in training.



Privacy Concern: When AI Starts Copy-Pasting

AI models sometimes do more than ***learn patterns***, they ***memorise*** *and* ***regurgitate*** their training data.

A 2020 study on GPT-2 found that around **0.1%** of its text outputs contained exact phrases **repeated** from its training data.

0.1%? that is nominal, until we realise that 0.1% of billions of tokens is a lot of potential secrets.

Why is it Important?
Imagine what could happen if they were trained on **private emails, medical notes, or internal chat logs**.
Your AI might start quoting your boss – or worse, your therapist.

From the study's analysis (**What gets memorised?**):

- **News headlines and error logs** (harmless gossip)
- **Licenses, URLs, and configuration files** (mildly awkward)
- **Contact details, code, and personal names** (privacy nightmare)

The issue is not that AI *learns* – it is that sometimes it **remembers too well**.
True privacy-preserving AI teaches models to generalise, not to become a parrot

AI Never Forgets... Even When It Should

Another type of content that models can memorise:

Personal information and copyrighted material!

For example, when researchers prompted GPT-3 with the start of *Harry Potter and the Philosopher's Stone*, it flawlessly reproduced **a full page of the book (around 240 words!)**

And GPT-2 once went a step further – it accurately recalled a real person's contact details (name, email, phone, and fax) just from a tiny prompt.

It was like the AI saying:

"***Oh, you mean Peter from Southampton? Here's his number!"***

Too Much Information: When Data Meets Data

This is what happens when you *think* data anonymisation is enough...

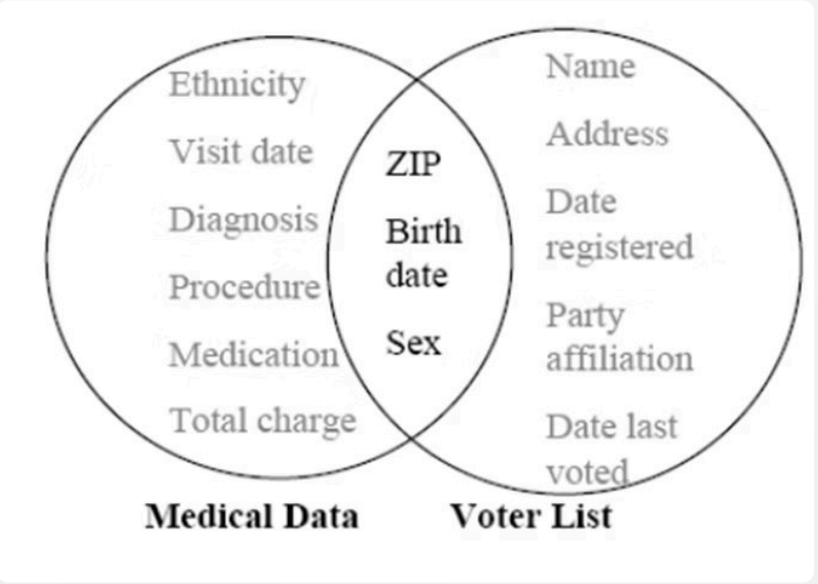
By linking two datasets; for example medical records and a voter list using shared details like **ZIP code, birth date, and gender**, it becomes surprisingly easy to identify individuals.

In a study, researcher *Latanya Sweeney* demonstrated how linking two “anonymous” datasets could expose personal identities.

She successfully re-identified the *Governor of Massachusetts* by matching de-identified *medical records* with publicly available *voter registration data*.

The surprising finding: **87% of the U.S. population can be uniquely identified using only three attributes — ZIP code, birth date, and gender.**

De-identified doesn’t always mean unidentified!



The Netflix Prize

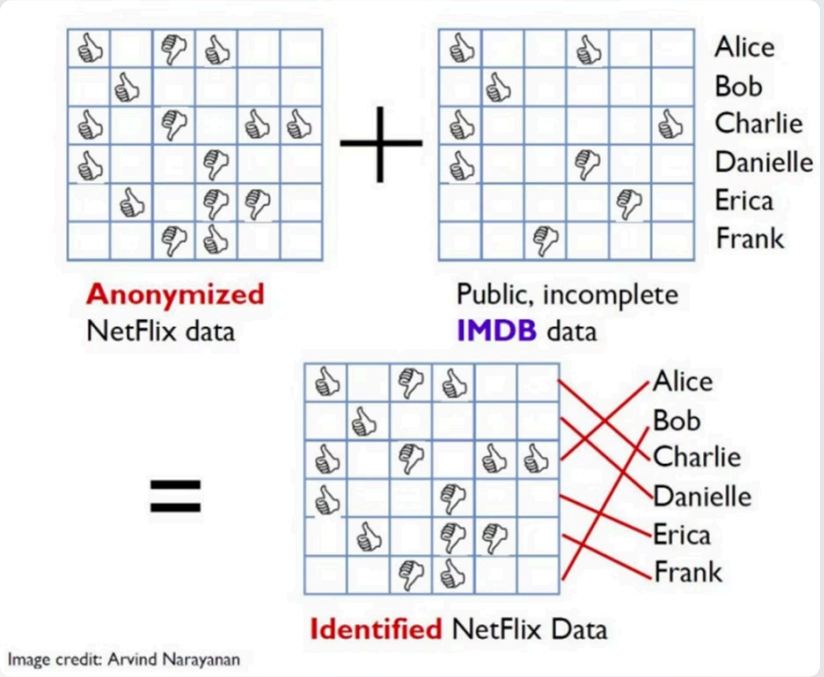
Between 2006 and 2009, Netflix launched a global challenge inviting researchers to improve its movie recommendation system. To help participants, Netflix released an “anonymised” dataset of users’ movie ratings.

However, researchers *Arvind Narayanan* and *Vitaly Shmatikov* later showed that the dataset was **not truly anonymous**.

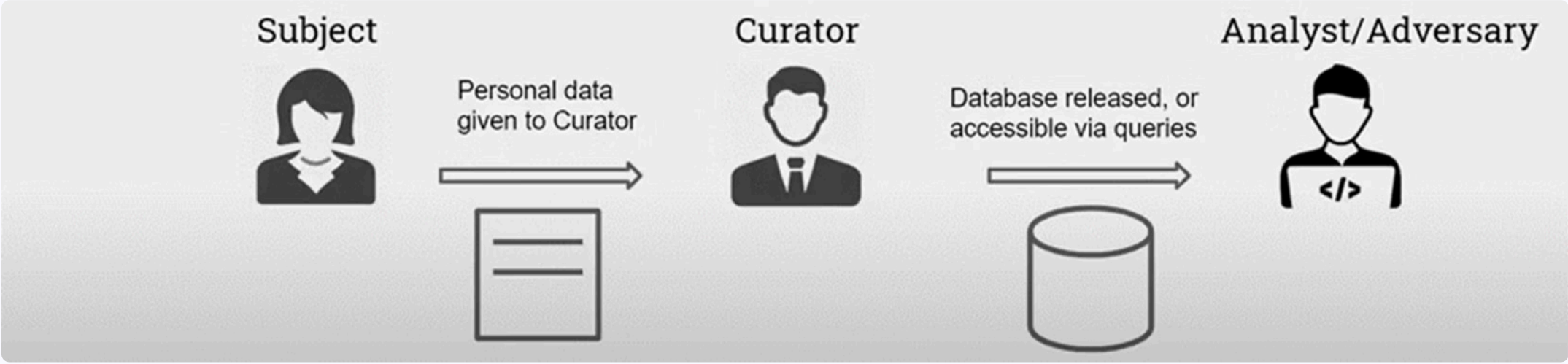
By cross-referencing the Netflix data with public ratings on **IMDb**, they could re-identify individual users

For example:

Someone rated the same movies around the same dates in both Netflix and IMDb. This was enough to *uniquely identify users* in the Netflix dataset, linking them to real IMDb accounts and in some cases, revealing personal or sensitive viewing habits.



Setting for Defining Privacy

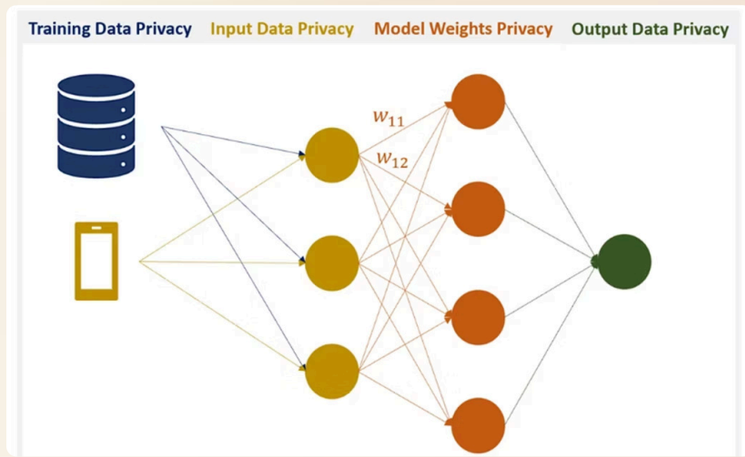


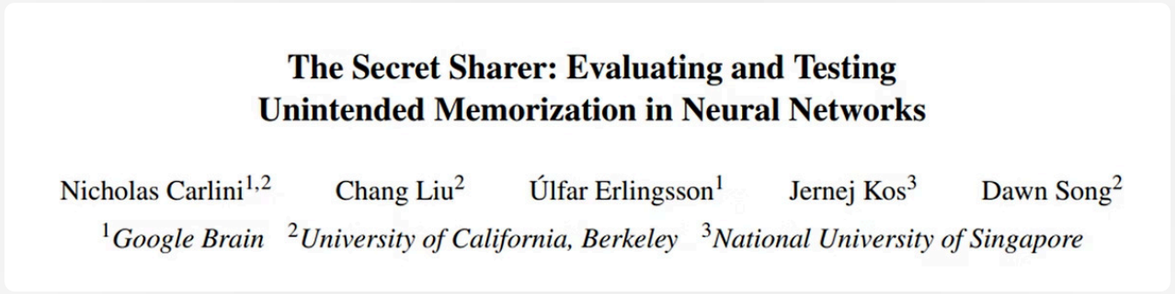
Privacy is a promise from the curator to the subject that no harm will come to them by including their data in the database.

The Four Pillars of Privacy-Preserving AI

Privacy-Preserving Machine Learning (PPML) protects sensitive data at every stage of an ML model – from input to output – ensuring that no private information is exposed or misused.

- **Training Data Privacy (Blue):**
  Protects the datasets used to train the model.
  Example: Federated Learning keeps raw user data on devices.
- **Input Data Privacy (Yellow):**
  Protects the data users send to a trained model.
  Example: Encrypted or masked data used during inference.
- **Model Weights Privacy (Orange):**
  Protects the learned parameters (w_{11}, w_{12}, etc.) so they cannot be reverse-engineered. **Example:** Homomorphic encryption to prevent model theft or reconstruction.
- **Output Data Privacy (Green):**
  Ensures the results do not leak information about the input or training data.
  Example: Differential Privacy adds controlled noise to outputs.





The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks

Nicholas Carlini[1,2] Chang Liu[2] Úlfar Erlingsson[1] Jernej Kos[3] Dawn Song[2]

[1]*Google Brain* [2]*University of California, Berkeley* [3]*National University of Singapore*

Training Data Privacy

The researchers wanted to see if **AI models accidentally memorise private information** from their training data.

So, they did a little experiment:

1. They took a normal text dataset called the *Penn Treebank Dataset* (a common dataset used to train language models).
2. They added a fake secret message to it – something that looked like a social security number:

"the random number is ooo-oo-oooo"

3. Then they trained a small text-generating AI model (one that learns to predict the next character or word).

4. They only used 5% of the dataset – so the AI had limited data to learn from.

After training, they tested the model to see what it remembered.

Result:
When the model was performing really well (meaning it had learned the training data accurately), it also remembered and repeated the fake number exactly when prompted with part of the phrase!

Training Data Privacy – Measuring Memorisation (Exposure Metric)

After Carlini et al. discovered that neural networks could memorise hidden secrets, they asked:

“Can we measure how likely it is that a model remembers and reveals a secret?”

To do this, they introduced a metric called **Exposure**.

For example: We plant “the secret code is ooo-oo-oooo”) inside the training data.
Then we check how easily the model can reproduce it when prompted.

Exposure Formula:
Exposure = How many guesses an attacker would need before correctly generating the secret.

- If exposure is **high** → the secret is **well hidden** (safe).
- If exposure is **low** → the secret is **easily recalled** (unsafe).
- So, **a smaller index = higher memorisation risk.**

Training Data Privacy - Aggregation of Teacher Ensembles (PATE – Google, 2018)

PATE is an **ensemble-based privacy technique** that trains multiple teacher models on disjoint (separate) subsets of sensitive data.

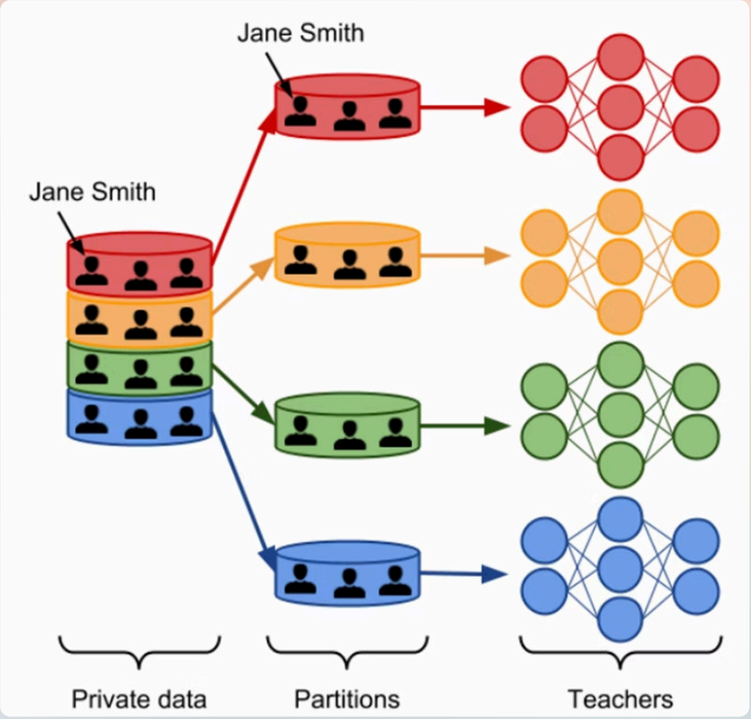
Each teacher learns independently, so no single model sees the entire dataset.

When making a prediction (inference), these teacher models **vote** on the answer.

- If most teachers agree, the output is safe to share, since it does not expose any individual's private data.
- The final **student model** learns from these aggregated votes – not from the raw private data.

Diagram:

- Each coloured block = a partition of private data (e.g., red, orange, green, blue).
- Each trained model = a *teacher*.
- The student model aggregates their outputs



Training Data Privacy - Differentially Private Stochastic Gradient Descent (DP-SGD)

When we train a deep learning model, it “learns” by looking at examples again and again, adjusting itself to make better predictions.

But if the model learns **too precisely**, it might accidentally remember details about specific people in the training data – like names, addresses, or medical results.

Differential Privacy (DP) solves this problem.

It ensures that the model learns **general patterns, not personal information.**

How it works?

- During training, each update to the model is slightly **blurred with random noise.**
- This noise hides the influence of any one person’s data.
- As a result, even if someone tries to inspect or hack the model, they cannot tell whether a particular person’s data was used.

It is just the regular “learning process” (Stochastic Gradient Descent) with an extra privacy layer added.

Input and Output Privacy

Both **what goes in** (the user's data) and **what comes out** (the model's results) must stay protected.
Only the user, and no one else, should be able to see or access them.

This ensures compliance with the **Four Pillars of Privacy-Preserving AI** – keeping the entire data lifecycle secure.

Why is it Important?

If input and output privacy are not maintained, private data can:

- **Be misused** - for example, *location tracking* scandals like those reported by *The New York Times*.
- **Fall into the wrong hands** - through hacks or data breaches.
- **Be used without consent** - like when *Amazon admitted that employees listened to Alexa recordings*.



Input and Output Privacy - Solutions

1. **Homomorphic Encryption (HE)** Allows computation directly on encrypted data without ever decrypting it. So, even if the data is processed by a third party, it remains private. **Example:** A hospital encrypts patient data before sending it to the cloud for AI analysis – the AI model can make predictions without ever seeing the raw data.
2. **Secure Multiparty Computation (MPC)** Multiple parties work together to compute a result without revealing their individual data. **Example:** Several banks jointly train a fraud detection model without sharing their customers' private transaction records.
3. **Federated Learning (FL)** The model travels to the data, not the other way around. Each device or institution trains locally and sends only the model updates (not data) back to a central server. **Example:** Google uses FL so your phone helps improve predictive text models without uploading your messages.

Model Privacy

Model privacy means protecting the trained machine learning model itself – not just the data used to train it.

A model can unintentionally leak information about the people or data it learned from, and it can also be stolen or reverse-engineered.

1. **Person-to-Person Similarities:** Two patients with almost identical heart rate and lifestyle data. If one patient's data becomes known, the model may reveal insights that indirectly expose the other.
2. **Support Vector Machines (SVMs):** SVMs separate data by drawing a boundary between different classes. In a medical model distinguishing "healthy" vs "unhealthy," the few patient records that sit closest to the dividing line are stored as support vectors. If exposed, these can be reverse-engineered to reconstruct the original private data.

Solutions and Trade-offs:

- **Differential Privacy** - Introduces random noise to prevent data leaks.
  Trade-off: Reduces accuracy if overused, as too much noise harms learning.
- **Homomorphic Encryption** - Keeps the model encrypted even during computation, allowing it to process data without ever being decrypted.
  Trade-off: Extremely computationally expensive.

There Is No Single Solution to Privacy

Perfectly privacy-preserving AI does **not** come from one technology alone.
Each method protects a *different pillar* – so we must **combine** them to cover all bases.

Think of it as building a strong fortress:

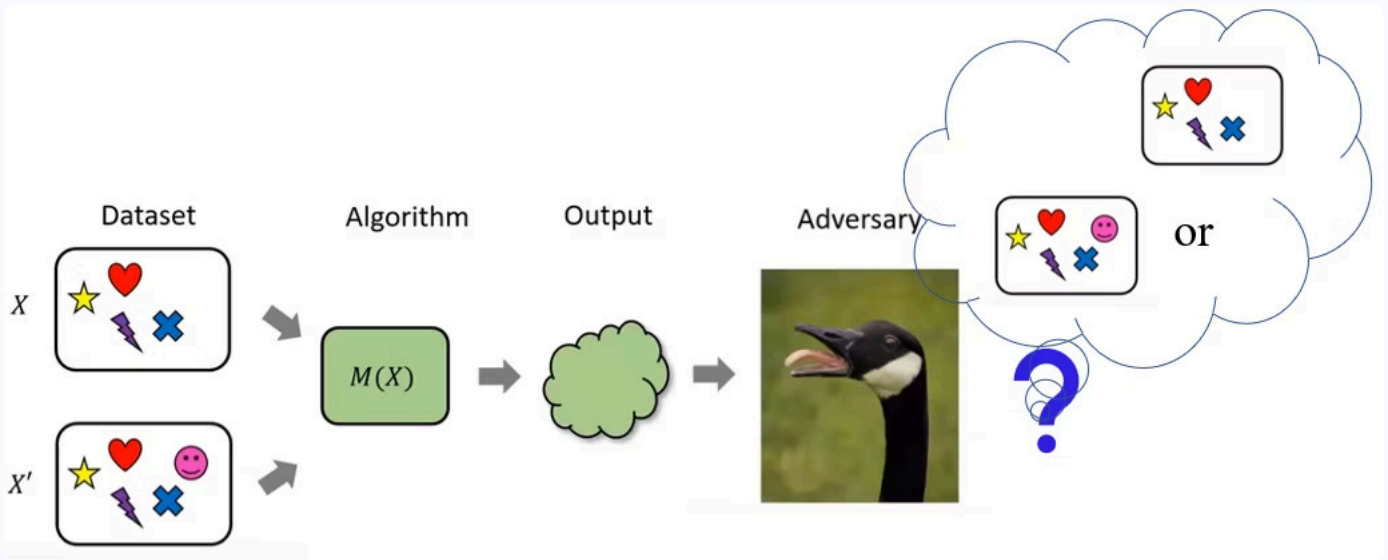
- Walls = **Encryption**
- Guards = **Differential Privacy**
- Secure Gateways = **MPC or Federated Learning**

Common Winning Combinations:

- **Homomorphic Encryption + Differential Privacy**
  → Protects *inputs, outputs, and training data*
- **Secure Multiparty Computation (MPC) + Differential Privacy**
  → Protects *data sharing and aggregation*
- **Federated Learning + Differential Privacy + MPC**
  → Protects *users' local data and model updates*
- **Homomorphic Encryption + PATE**
  → Protects *teacher models and student outputs*
- **Secure MPC + PATE**
  → Enables *safe multi-institution collaboration*
- **Federated Learning + PATE + Homomorphic Encryption**
  → Full-stack privacy – *end-to-end protection* across users, data, and mode

Privacy is teamwork — not a solo act!

Differential Privacy



Differential Privacy (DP) means the model or algorithm does not “notice” if one person’s data is in the dataset or not.
So, even if you remove or add one individual’s record, the final output (the model’s prediction, report, or statistics) will stay *almost the same*.

“Your data could be in there or not – but nobody can tell the difference from the outside.”

If someone (**the adversary**) tries to figure out whether your personal data was used in training, DP ensures they can never be sure.
For example:

- Suppose a hospital publishes “average blood pressure” from patient data.
- If your data were included, the result should be almost identical to when it was not.

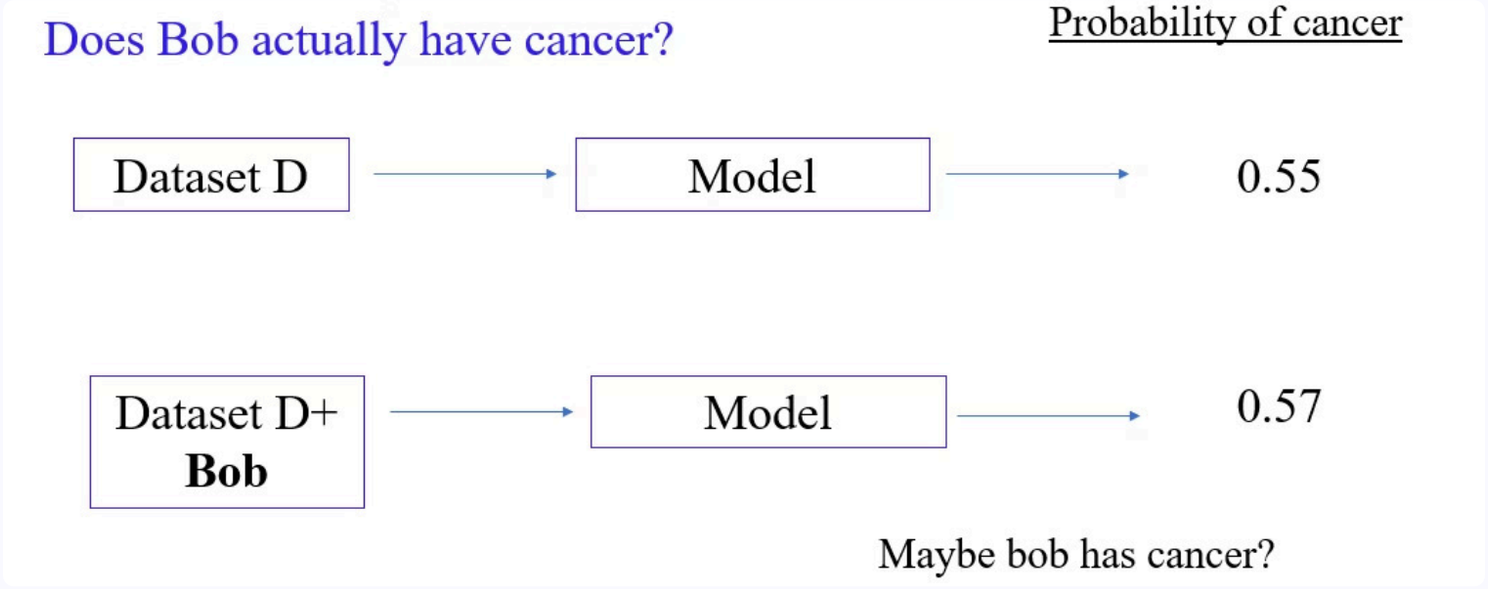
Left: Two datasets – one with the pink smiley (your data) and one without.

Middle: The algorithm processes them both and produces almost the same “output cloud.”

Right: The **adversary** is confused – they cannot tell which dataset was used.

Plausible Deniability — The Heart of Differential Privacy

Differential Privacy gives every individual in the dataset “plausible deniability.”



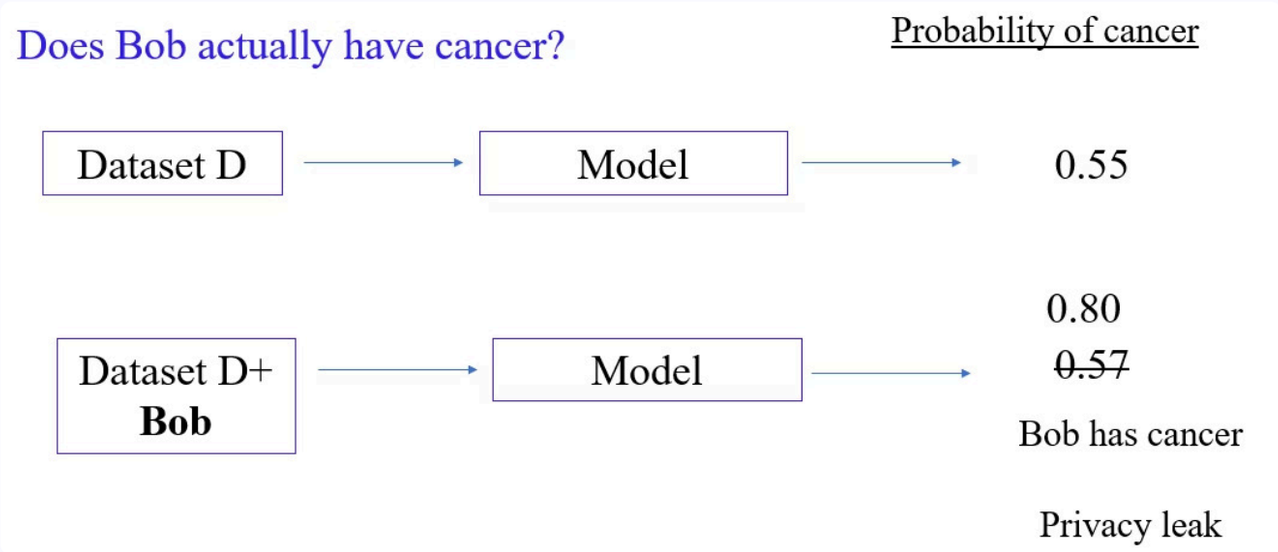
Example:

- The model predicts the probability of cancer as **0.55** (without Bob) and **0.57** (with Bob).
- The change is *tiny* – too small to say whether Bob’s record affected the result.
- So, if someone asks: “Does Bob have cancer?” the model output doesn’t reveal it clearly.

Because every individual’s contribution has minimal effect on the output, Bob can **plausibly deny** that his data influenced the result.

When Differential Privacy Fails — Privacy Leak Example

If adding *just one person's data* (like Bob's) changes the model output significantly, it means the model is **not differentially private — and privacy has been compromised.**

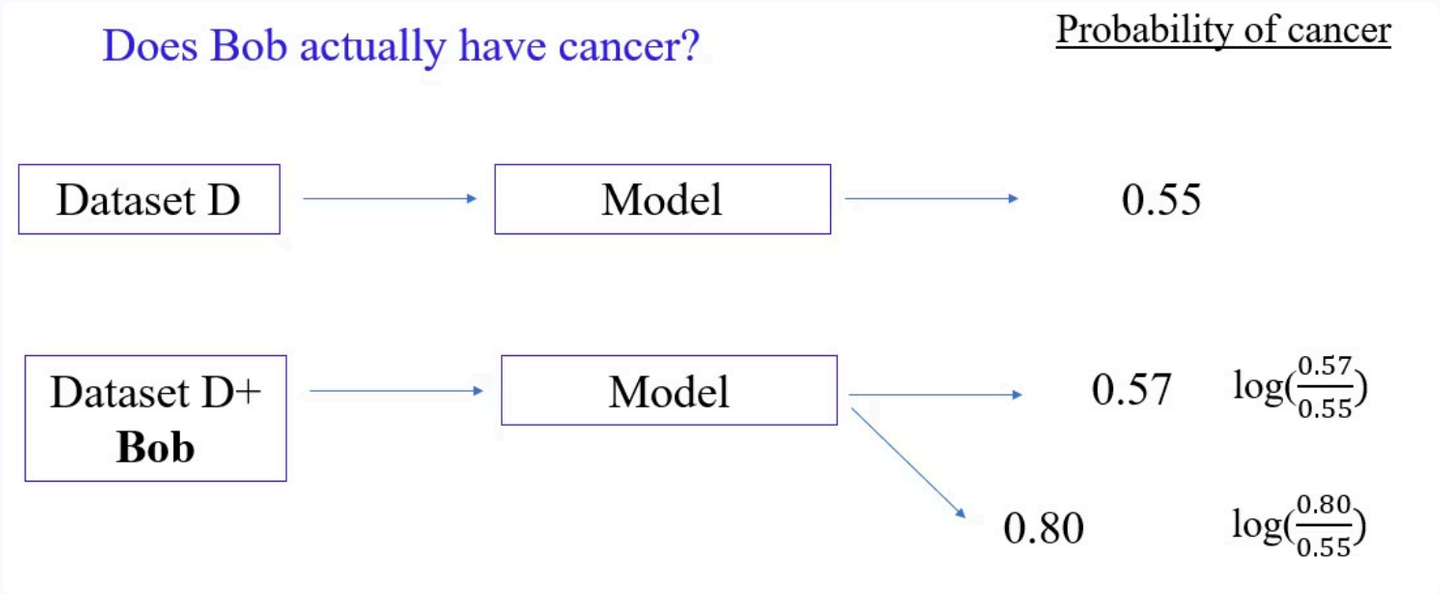


- In the first case, the model trained on Dataset D gives a **0.55** probability of cancer.
- When Bob's record is added (Dataset D + Bob), the probability jumps to **0.80**.
- This big change reveals that Bob's data has a strong influence – suggesting Bob *has cancer*.
- This violates the principle of *plausible deniability*.
- The adversary (attacker) can infer private information about Bob just by comparing outputs.

Measuring Privacy — The Role of ε (Epsilon)

In Differential Privacy, the parameter **ε (epsilon)** quantifies *how much* the output of a model can change when one person's data (like Bob's) is added or removed.

It measures the privacy loss — smaller ε means stronger privacy.



Mathematically:

We compare the two outputs (with and without Bob) using a **log ratio**:

- When this ratio ≈ 1 (or log ratio ≈ 0)→ Bob's data has little effect → Strong privacy
- When the ratio is much larger→ Bob's data strongly influences the result → Weak privacy

Differentially Private Mechanisms - Laplace

Suppose, we have a list of people. Each person's data is **1** if they smoke, and **0** if they do not.
We want to know how many people smoke in total – but we do not want anyone to know whether a specific person (like Bob) smokes.

Problem: If we simply count the smokers and publish the total, someone could notice small changes and figure out if Bob's data was added or removed.

For example:

- Without Bob → 45 smokers
- With Bob → 46 smokers
  That's a clear privacy leak.

The Solution - Add Noise

Instead of giving the exact number, we add a bit of **random noise** to the answer.
This noise comes from something called the **Laplace distribution**,
which makes the number "fuzzy" – slightly up or down.

So instead of "46," the result might be **45.8** or **47.2**.
Now, nobody can tell if Bob was in the dataset or not.

The Privacy Control (ε - epsilon)

- **ε (epsilon)** controls how much noise is added.
  - **Small ε → more noise → stronger privacy**
  - **Large ε → less noise → weaker privacy but more accurate results**

Summary

- What is privacy in ML
- What are four pillars of Perfectly-Privacy Preserving AI
- Differential privacy in ML
- How to implement the differential privacy (DP) principles in existing ML algorithms to provide privacy guarantees for training data using TensorFlow Privacy

Questions

